

Profiling protokol

Tým: audentes Fortuna iuvat

Vstup o velikosti 10^6

Call graph

Call graph					
index	% time	self	children	called	name
<spontaneous>					
[1]	100.0	0.01	0.02		main [1]
		0.01	0.00	2000000/2000000	addition(double, double) [2]
		0.01	0.00	1000001/1000001	exponentiation(double, double) [3]
		0.00	0.00	3/3	multiplication(double, double) [10]
		0.00	0.00	2/2	division(double, double) [12]
		0.00	0.00	2/2	subtraction(double, double) [11]
		0.00	0.00	1/1	nthRoot (double, double) [13]

[2]	33.3	0.01	0.00	2000000/2000000	main [1]
		0.01	0.00	2000000	addition(double, double) [2]

[3]	33.3	0.01	0.00	1000001/1000001	main [1]
		0.01	0.00	1000001	exponentiation(double, double) [3]

[10]	0.0	0.00	0.00	3/3	main [1]
		0.00	0.00	3	multiplication(double, double) [10]

[11]	0.0	0.00	0.00	2/2	main [1]
		0.00	0.00	2	subtraction(double, double) [11]

[12]	0.0	0.00	0.00	2/2	main [1]
		0.00	0.00	2	division(double, double) [12]

[13]	0.0	0.00	0.00	1/1	main [1]
		0.00	0.00	1	nthRoot (double, double) [13]

Flat profile

Each sample counts as 0.01 seconds.						
% time	cumulative seconds	self seconds	calls	self ns/call	total ns/call	name
33.33	0.01	0.01	2000000	5.00	5.00	addition(double, double)
33.33	0.02	0.01	1000001	10.00	10.00	exponentiation(double, double)
33.33	0.03	0.01				main
0.00	0.03	0.00	3	0.00	0.00	multiplication(double, double)
0.00	0.03	0.00	2	0.00	0.00	subtraction(double, double)
0.00	0.03	0.00	2	0.00	0.00	division(double, double)
0.00	0.03	0.00	1	0.00	0.00	nthRoot (double, double)

Vstup o velikosti 10^3

Call graph

Call graph					
granularity: each sample hit covers 4 byte(s) no time propagated					
index	% time	self	children	called	name
		0.00	0.00	2000/2000	main [6]
[8]	0.0	0.00	0.00	2000	addition(double, double) [8]

		0.00	0.00	1001/1001	main [6]
[9]	0.0	0.00	0.00	1001	exponentiation(double, double) [9]

		0.00	0.00	3/3	main [6]
[10]	0.0	0.00	0.00	3	multiplication(double, double) [10]

		0.00	0.00	2/2	main [6]
[11]	0.0	0.00	0.00	2	subtraction(double, double) [11]

		0.00	0.00	2/2	main [6]
[12]	0.0	0.00	0.00	2	division(double, double) [12]

		0.00	0.00	1/1	main [6]
[13]	0.0	0.00	0.00	1	nthRoot (double, double) [13]

Flat profile

Each sample counts as 0.01 seconds.							
no time accumulated							
% time	cumulative seconds	self seconds	calls	self Ts/call	total Ts/call	name	
0.00	0.00	0.00	2000	0.00	0.00	addition(double, double)	
0.00	0.00	0.00	1001	0.00	0.00	exponentiation(double, double)	
0.00	0.00	0.00	3	0.00	0.00	multiplication(double, double)	
0.00	0.00	0.00	2	0.00	0.00	subtraction(double, double)	
0.00	0.00	0.00	2	0.00	0.00	division(double, double)	
0.00	0.00	0.00	1	0.00	0.00	nthRoot (double, double)	

Vstup o velikosti 10

Call graph

Call graph (explanation follows)					
granularity: each sample hit covers 4 byte(s) no time propagated					
index	% time	self	children	called	name
		0.00	0.00	20/20	main [6]
[8]	0.0	0.00	0.00	20	addition(double, double) [8]

		0.00	0.00	11/11	main [6]
[9]	0.0	0.00	0.00	11	exponentiation(double, double) [9]

		0.00	0.00	3/3	main [6]
[10]	0.0	0.00	0.00	3	multiplication(double, double) [10]

		0.00	0.00	2/2	main [6]
[11]	0.0	0.00	0.00	2	subtraction(double, double) [11]

		0.00	0.00	2/2	main [6]
[12]	0.0	0.00	0.00	2	division(double, double) [12]

		0.00	0.00	1/1	main [6]
[13]	0.0	0.00	0.00	1	<u>nthRoot</u> (double, double) [13]

Flat profile

Each sample counts as 0.01 seconds.							
no time accumulated							
% time	cumulative seconds	self seconds	calls	self Ts/call	total Ts/call	name	
0.00	0.00	0.00	20	0.00	0.00	addition(double, double)	
0.00	0.00	0.00	11	0.00	0.00	exponentiation(double, double)	
0.00	0.00	0.00	3	0.00	0.00	multiplication(double, double)	
0.00	0.00	0.00	2	0.00	0.00	subtraction(double, double)	
0.00	0.00	0.00	2	0.00	0.00	division(double, double)	
0.00	0.00	0.00	1	0.00	0.00	<u>nthRoot</u> (double, double)	

Shrnutí

Vzorec, který byl využit nejčastěji volá addition a multiplication, což je způsobeno tím, jak je naimplementováno načítání čísel. Tudíž právě to, jak jsou čísla načítána je místo v programu, kde je smysluplné se zaměřit na optimalizaci.

Na profilování byl využit program „GNU gprof (GNU Binutils for Ubuntu) 2.42“, který měří po milisekundách a jak je vidět, tak při profilování na vstupech o velikosti 10 a 10^3 nebyl ani schopen moc naměřit, jelikož program skončil rychleji, než stihla vůbec milisekunda uběhnout.

Měření bylo provedeno na stroji s následujícími specifikacemi:

CPU	Intel Core i7-11700F (11. generace), 8 jader / 16 vláken, 2.50 GHz
RAM	16 GB (dostupných 14 GB)
Architektura	x86_64 (64bit)
OS	Ubuntu 24.04.2 LTS (běží pod WSL2, jádro 5.15.167.4-microsoft-standard-WSL2)
Kompilátor	g++ 13.3.0 (Ubuntu)
Volby překladače	-pg